

Assessment Tool-3
Co-1
Technical Presentation

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Course Code:- ITA0612

Course Name:- Machine Learning.

Learning Problems and Perspectives

machine learning enables computers to learn patterns from data and make predictions or decisions without being explicitly.

Types of learning:-

1. Supervised learning - learns from labelled data.

Ex:- disease classification.

2. Unsupervised learning:- Finds patterns in unlabelled data.

Ex:- customer clustering.

3) Reinforcement learning:- learns through rewards and penalties. Ex:- game-playing robots.

Perspectives:-

→ Symbolic approach:- Represents knowledge using rules and logical structures

→ Statistical approach:- Uses mathematical and probabilistic models to learn patterns

Major issues:- Poor data quality, overfitting, insufficient training data, and high computational complexity.

2) Concept learning and Hypothesis Representation:-

Concept learning is the Process of learning a General concept from a set of trainings

Ex:- To identify whether a day is suitable for Playing.

Weather $\rightarrow$ Play

A hypothesis may be:

Sunny & warm $\rightarrow$ Play

Hypotheses can be arranged from general to specific.

Concept - learning Process:-

1. Collect training examples
 2. Represent the examples using attributes.
 3. Generate Possible hypotheses.
 4. Compare hypotheses with training examples.
 5. Refine the hypothesis using Positive and negative examples.
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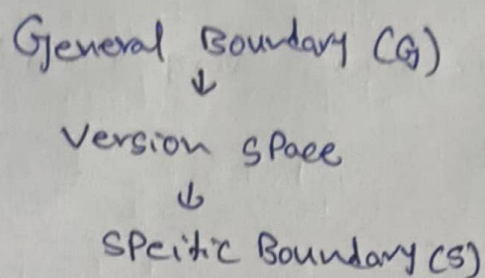
3) Version Spaces and Candidate Elimination:

A version space is the set of all hypotheses that are consistent with the available training examples.

The candidate elimination algorithm maintains two boundaries:-

- S - Specific boundary: most specific consistent hypotheses.
- G - General boundary: most general consistent hypotheses.

Working:-



For a Positive example, hypotheses that do not cover the example are removed, and S is generalized when necessary.

For a negative example, hypotheses that cover the example are removed and G is specialized when necessary.

4) Inductive Bias in learning Algorithms.

Inductive bias is the set of assumptions made by a learning algorithm to generalize from limited training data.

There are two important types:-

1. Restriction bias:- Restricts the possible hypotheses that can be considered.

2. Preference bias:- Allows many hypotheses but prefers some hypotheses over others.

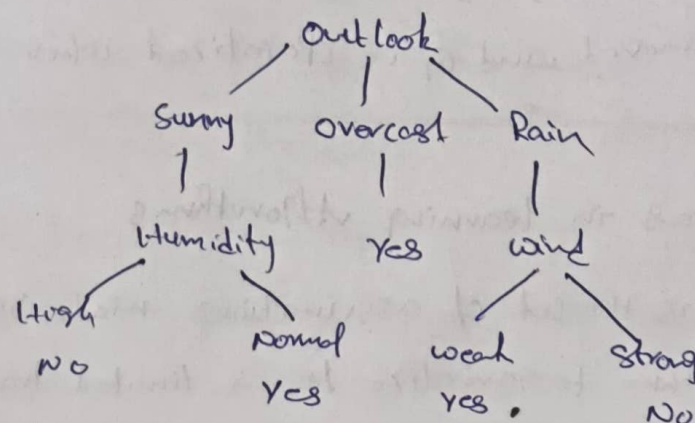
Importance:-

- Helps the algorithm generalize to unseen data.
- Reduces the search space.
- Influences hypothesis selection.
- Makes learning computationally practical.

For example, a decision tree algorithm may prefer a smaller tree over a very complex tree.

5) Decision Tree learning and Representation:-

A decision tree represents knowledge in the form of a tree structure.



Important Concepts:-

Entropy:- Measures the impurity or uncertainty in the data.

Information Gain:- Measures how well an attribute separates the data.

Tree Pruning:- Removes unnecessary branches to reduce overfitting.

Advantages:-

- Easy to understand.
- simple to interpret.
- useful for classification and decision making.

Limitations:-

- small changes in data can change the tree.
- large trees may become complex.